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Communicating trust and trustworthiness through scientists' biographies: Benevolence beliefs

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Abstract

A goal of many science communicators is to foster trust in scientists and their work. This study investigates if existing textual resources that scientists create in the course of their regular activities can be improved to enhance perceptions of scientists as trustworthy. Building on Mayer et al.'s integrative model of organizational trust, we examine how communicating benevolence through short biographies can affect trustworthiness perceptions using a 3 (degree of benevolence information: high, unspecified, low) × 3 (research area: crop genetics, corn and soy genetics, biotechnology use) survey design. We find that the degree of benevolence information significantly influences perceptions of benevolence and integrity, as well as willingness to trust, with these effects being consistent across different research areas. However, the degree of benevolence communicated had no significant effect on the perceived competence of the scientists. These findings underscore the importance of highlighting benevolence in communication to positively influence trustworthiness perceptions, thus offering insights for science communication practices.

Keywords

benevolence, goodwill, science communication, trust, trustworthiness, warmth

Scientists want people to turn to them for advice (Besley et al., 2020), yet accepting advice inherently requires making oneself vulnerable to the one giving the advice. The willingness to make oneself vulnerable is a classic definition of behavioral trust in the integrative model of organizational trust, a well-established effort to consolidate literature on trust and trustworthiness from across the social sciences (Schoorman et al., 2007). Some recent efforts to measure perceptions of

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scientists' trustworthiness have used the integrative model of trust (Besley et al., 2021; Hendriks et al., 2015), but further investigation is needed into how these perceptions form.

The present study examines the efficacy of using short scientist biographies as a way to foster benevolence beliefs and willingness to trust scientists. These biographies, succinct informational text pieces, are often used to introduce scientists across official and personal channels. Information from such biographies can be readily reproduced or referenced in other channels (e.g. talks, poster presentations, press articles, and online search engines). Given their simplicity and prevalence, biographies could serve as ideal boundary objects, demonstrating how strategic communication choices can enhance the public's perceptions of scientists, even if only in the short term (Bevan et al., 2020).

We use the "integrative model" from the organizational psychology approach as it (a) consolidates elements from a broad range of literature on trust and (b) differentiates between trustworthiness *perceptions* (benevolence, integrity, ability) and trusting *behavioral intention* as the willingness to make oneself vulnerable as a function of these perceptions. Specifically, Mayer et al. (1995) conceptualize benevolence as the trust giver's (i.e. the trustor) belief that the trust receiver (i.e. the trustee) is likely "to want to do good" or be positively oriented to the trustor. Integrity, in contrast, focuses on perceptions about a trustee's sense of morality and justice (i.e. "a set of principles that the trustor finds acceptable"), while ability involves the trustor's perception of necessary skills and competence needed in the relevant domain (Mayer and Davis, 1999: 124). Given that messages or behaviors can communicate information about motivations without necessarily communicating information about integrity, there are sound conceptual and theoretical reasons to differentiate perceptions of motivation (i.e. "benevolence") from perceptions of integrity (Besley et al., 2021).

A belief is an estimate of the likelihood that a certain piece of information is correct, computed based on previously acquired knowledge (Wyer and Albarracín, 2005). As caring for others is generally viewed as virtuous, stating (or agreeing with a statement) that a scientist cares about others can be understood as an *evaluative belief* containing both a statement about perceived reality and a normative judgment (Fishbein and Ajzen, 2010). The reported intention toward a scientist can then be understood as the sum of evaluative beliefs that are *salient* (i.e. accessible in memory) to that intention. The assumption underlying this work is that exposure to salient information is relevant to fostering changes in estimating information, affect short-term perceptions of trustworthiness, and ultimately influence willingness to trust, such that:

Exposure to benevolence information	→	New knowledge informs estimation of information	→	Perceived benevolence	→	Willingness to trust

There is a reasonable basis to expect that conveying some types of information could foster beliefs about scientists' benevolence. For instance, previous work on communicating benevolence-type perceptions (e.g. "warmth" or "goodwill") about scientists has shown that prosocial quotations (Zahry Nagwan and Besley, 2021b) and non-verbal cues such as smiling or working in groups (Barger and Grandey, 2006; Zahry Nagwan and Besley, 2021a) can foster perceived warmth of scientists. In medical contexts, patients and caregivers have a more positive view of physicians who demonstrate caring behaviors (Sharp et al., 1992), and polite communication styles can enhance the perceived likability of a communicator (Yuan et al., 2019).

Our experiment investigates the effect of using short scientist biographies to communicate benevolence information about a scientist on respondents' short-term benevolence perceptions about a scientist. If such benevolence information about a scientist is salient to trustworthiness beliefs about that scientist in the short term, there should be a significant difference in the trustworthiness perceptions between participants exposed to benevolence-emphasized profiles and those who are not (Zaller, 1992). We also expect that benevolence information will have an effect on respondents' willingness to make themselves vulnerable to (i.e. willingness to trust) the scientist's advice. Our hypotheses are as follows:

Hypothesis 1 (H1). Emphasizing benevolence information in a scientist's profile will lead to higher ratings of a scientists' perceived benevolence, compared to profiles that do not emphasize benevolence information.

Hypothesis 2 (H2). Emphasizing benevolence information in a scientist's profile will lead to higher ratings of willingness to trust a scientist, compared to profiles that do not emphasize benevolence information.

Hypothesis 3 (H3). Perceived benevolence will positively mediate the relationship between degree of benevolence information and willingness to trust a scientist.

We focus on benevolence in the context of willingness to trust for three reasons. First, scientists are already viewed as highly competent by majority of people, both in the United States (e.g. Funk et al., 2019) and globally (see Online Appendix for: Gallup, 2019). Therefore, the potential positive movement might be more significant in the areas of benevolence and integrity. Second, people appear to weigh benevolence judgments more heavily than competence judgments when determining trustworthiness (i.e. deciding whether others *want* to hurt them seems to be of more immediate concern than deciding if they *can* in many contexts) (Fiske et al., 2007; Wojciszke and Abele, 2008). For instance, support for research into risk-related technologies such as nuclear energy and genetically modified food is more highly correlated with benevolence-related beliefs than competence-related beliefs (Besley et al., 2021). Third, it may be more straightforward to communicate benevolence over integrity: Crafting statements about a scientist's motivations for research may be simpler than describing their set of principles. Communicating prosocial motivations seems suited to generating concrete, positive messages (e.g. "I <u>do</u> my research to help"), while communicating integrity often leads to defensive statements (e.g. "I <u>do not</u> take money from industry"), though positive integrity messages can be made possible (e.g. "I make all my data publicly accessible").

Finally, while our formal hypotheses focus on benevolence perceptions and willingness to trust, we also propose the following research questions to investigate the effect of communicating benevolence information on other dimensions of trustworthiness:

Research Question 1 (RQ1). What effects does benevolence information have on perceived competence and willingness to trust the scientist?

Research Question 2 (RQ2). What effects does benevolence information have on perceived integrity and willingness to trust the scientist?

1. Method

We conducted a 3×3 between-subjects experiment, manipulating three levels of communicated benevolence (high, unspecified, and low) across three different research areas (crop genetics, corn and soy genetics, biotechnology use). Our primary analyses focus on the effect of the degree of benevolence (high, unspecified, and low) on perceptions of benevolence (H1 and H3), trust as willingness to be vulnerable (H2 and H3), and other trustworthiness dimensions (RQ1 and RQ2).

We also test to see if effects vary across different research areas to ensure that any effects found occur consistently across several contexts.

Stimuli development

We modeled fictional profiles after publicly viewable biographies of scientists at land-grant universities who were included on a list of highly cited researchers in the physical and social sciences. Based on pretests reported in previous research on conflict-of-interest perception (Besley et al., 2017), the scientist was described as a professor at the College of Agriculture at Purdue University, a land-grant university with a name that possessed no explicit regional affiliation. This information was kept consistent across profiles we used for stimuli, manipulating only the details of the fictional scientist's research and activities. The topic of genetically engineered food provided a reasonable context in which to explore science-related communication effects for this study. It has the advantage of remaining relatively apolitical, such that the effects of benevolence information would not be overwhelmed by ideological factors largely outside of communicators' control (Kahan, 2015). This issue is also relevant to science and risk communication research and the broader scientific community (National Academies of Sciences, Engineering, and Medicine et al., 2016).

Independent variables

Degree of benevolence. To develop the stimuli used for our benevolence manipulations, we recruited Amazon Mechanical Turk workers, ensuring that none of them participated in the main study, to evaluate various statements about research orientation, impact, and engagement with the community. Participants were asked to evaluate 15 statements (e.g. "The scientist often provides scientific advice to farmers") on a five-point, item-specific scale taken from our five-item measure of benevolence derived from previous work (Besley et al., 2020; Hendriks et al., 2015). These statements were the basis for the development of the biographies used for our experimental stimuli (see Online Supplemental Appendix A for pretest results).

Based on the ratings from our pretest, we developed three degrees of emphasis on benevolence information presented in scientist profiles. High-benevolence profiles emphasized research outcomes for the scientist's community and the environment (e.g. "identify environmentally friendly ways of growing plants that use less water and fewer chemicals"). Unspecified profiles omitted information about who the research benefited, used neutral language, or focused on scientific outcomes (e.g. "address the needs of a range of stakeholders"). Low-benevolence profiles highlighted research outcomes favoring private industry (e.g. "goal is to find new applications for the biomedical industry"). We operationalized high benevolence as research conducted for explicitly prosocial reasons and low benevolence research as research primarily driven by economic motives. While focusing on industry in low-benevolence might introduce a confound, the alternative—depicting trivial prosocial benefit—seemed less realistic. A neutral condition was also included as a control.

Research areas. We tested the effects of manipulated benevolence information with biographies of scientists in different research areas to ensure that our reported findings were a function of the substantive benevolence information in each biography and not merely attributable to the specific messages or the scientists' specific expertise (Brashers and Jackson, 1999). Research area 1 (crop genetics) described the scientist's research as related to the genetics of agricultural crops, dealing with threats, and acting as a scientific consultant. Research area 2 (corn and soy genetics) indicated that the scientist's research involved modifying the genes of corn and soy, developing new varieties of crops, and serving as a scientific advisor. Finally, Research area 3 (biotechnology use) indicated that the scientist's research focused on using biotechnology in agriculture, the genetic factors that make plants suitable for different regions, and the provision of scientific advice to groups or institutions.

Experimental design

We recruited a sample of Amazon Mechanical Turk Workers to read and evaluate one of the nine short fictitious biographies about a plant biologist. Each participant was randomly assigned to view and rate a single scientist profile out of the nine available. These profiles were created from combinations of three degrees of benevolence (high, unspecified, and low) with three distinct research areas (crop genetics, corn and soy genetics, biotechnology use). Participants were instructed to spend at least 1 minute reading the profile (enforced using the Qualtrics survey timer) and then complete the survey instrument. The instrument included a five-item measure of perceived benevolence (similar to the one used in the pretest), questions on willingness to trust (three items), perceived competence (four items), and perceived integrity (four items). All questions were asked using five-point, item-specific scales (see Table 1). The survey asked participants to provide their demographic information and included a measure of science literacy, the "Oxford scale" adapted from Allum et al. (2008).

We restricted the participation to workers with an approval rating of 95% and above, which is considered a sufficient condition for high data quality (Peer et al., 2014). To verify that the respondents read the biography and correctly identified the profession of the scientist, we implemented two treatment checks. We also implemented two stimulus checks to verify whether they perceived the manipulation of the degree of benevolence. Responses were removed if they failed both treatment and stimulus checks. Several participants who were assigned to the unspecified condition correctly answered questions about the scientist's university and what profession they just read about, but fewer than half correctly responded when asked information about which groups the scientist engaged with and what the goal of their work was. As the question wording about their perceptions of the scientist's intentions was deliberately ambiguous, it seems likely that a number of the participants resorted to guessing. This is between the number of respondents needed to find a medium and small effect size with an error probability of .05 and a power of .80 for both main and interaction effects according to G*Power estimates for analysis of variance (ANOVA) tests. To ensure the efficacy of random assignment after excluding responses that did not meet our criteria, we conducted an equality of proportions test. We found no significant differences in the baseline characteristics between the groups, indicating that the randomization was successful. Details regarding the test statistics and the distribution of participants by age, gender, education, and science literacy scores can be found in Online Supplemental Appendix C.

2. Results and data analysis

Factorial analysis of variance was used to test whether the degree of emphasis on benevolence information significantly affected trust, perceived benevolence, perceived competence, or perceived integrity between different degrees of benevolence (Table 2). Tukey honestly significant difference (HSD) post hoc comparisons were conducted to determine the significance of degree of benevolence and to verify whether there were significant differences between research areas.

Table 1. Aggregated descriptive statistics and mean comparisons for individual and summary variables by degree of benevolence information.

	Degree of benevolence information					
	High		Unspecified		Low	
	M	*SD*	*M*	*SD*	*M*	*SD*
Perceived benevolence (5 items) (*alpha* = *.90*)	4.45[a]	.59	4.19[b]	.72	3.94[c]	.74
. . . how much would you expect this scientist cares or does not care about helping others? . . .	4.35[a]	.81	4.13[a]	.79	3.80[b]	.84
. . . how much would you expect this scientist to be interested or uninterested in improving other people's lives? . . .	4.57[a]	.68	4.27[b]	.91	4.10[c]	.90
. . . how much would you expect this scientist to be considerate of others' needs? . . .	4.38[a]	.73	4.06[b]	.87	3.74[c]	.84
. . . how much would you expect this scientist to be concerned about the well-being of others? . . .	4.40[a]	.78	4.14[b]	.82	3.95[b]	.90
. . . how much would you expect this scientist to try to help others? . . .	4.54[a]	.69	4.34[a]	.84	4.09[b]	.84
Willingness to trust (2 items) (*alpha* = *.74*)	4.39[a]	.69	4.27[a]	.72	4.06[b]	.83
. . . how willing or unwilling would you be to take advice from this scientist in their area of expertise?	4.61[a]	.66	4.53[a]	.74	4.41[a]	.81
. . . to what degree should government decision-makers ask for advice from a scientist like this when making decisions in this area?	4.16[a]	.92	4.01[a]	.89	3.72[b]	1.02
Perceived competence (4 items) (*alpha* = *.83*)	4.77[a]	.42	4.73[a]	.48	4.72[a]	.48
. . . how intelligent or unintelligent do you think this scientist would be? . . .	4.84[a]	.45	4.80[a]	.58	4.76[a]	.48
. . . how skilled or unskilled do you think this scientist would be when it comes to conducting scientific research? . . .	4.78[a]	.56	4.75[a]	.55	4.71[a]	.55
. . . how knowledgeable do you think this scientist would be when it comes to conducting scientific research? . . .	4.78[a]	.58	4.72[a]	.66	4.78[a]	.51
. . . how much expertise do you think this scientist has when it comes to conducting scientific research? . . .	4.69[a]	.52[a]	4.66[a]	.57	4.63[a]	.70
Perceived integrity (4 items) (*alpha* = *.83*)	4.42[a]	.68	4.33[a]	.74	4.02[b]	.75
. . . how honest or dishonest do you think this scientist would be? . . .	4.45[a]	.72	4.39[a]	.79	4.15[b]	.82
. . . how ethical or unethical do you think this scientist would be? . . .	4.51[a]	.86	4.38[a]	.87	4.03[b]	.92
. . . how sincere or insincere do you think this scientist would be? . . .	4.39[a]	.83	4.28[ab]	.91	4.06[b]	.94
. . . how likely or unlikely is that it that this scientist would try to mislead others? . . .	4.34[a]	.99	4.26[a]	.98	3.87[b]	1.07
Total valid responses per condition	114		141		118	

Note. As there was no significant difference between research areas in the same condition ($p > .05$), the means presented in this table were obtained by combining all responses in each condition from research area 1 to 3; means, standard deviations, and *n* for each research area are provided in Online Supplemental Appendix E. All questions started with the subordinate clause "Based on the biography you just read" and ended with the phrase "Do you think they would be . . ." followed by a five-point item-specific scale that typically went from a "very" low level of perception to a "very" high level of perception (see Online Supplemental Appendix I). Alpha stands for Cronbach's alpha. Values of *n* ranged from 114 (high), 140–142 (unspecified), and 117–118 (low). Values in the same row without a common superscript (a, b, c) are significantly different ($p < .05$, Tukey's test).
SD: standard deviation.

Table 2. Analysis of variance for the effect of research area and degree of benevolence on willingness to trust, perceived benevolence, perceived competence, and perceived integrity.

Source	*df*	*SS*	*MS*	*F*	*p*	*eta*^{2}
Perceived benevolence (H1)						
Research area	2	1.16	.58	1.21	.30	.01
Degree of benevolence	**2**	**14.94**	**7.47**	**15.62**	**.00**	**.08**
Research area × Degree	4	1.18	.30	.62	.65	.01
Willingness to trust (H2)						
Research area	2	1.17	.59	1.05	.35	.01
Degree of benevolence	**2**	**6.18**	**3.09**	**5.51**	**.00**	**.03**
Research area × Degree	4	1.74	.44	.78	.54	.01
Perceived competence (RQ1)						
Research area	2	.28	.14	.67	.51	.00
Degree of benevolence	2	.20	.10	.47	.62	.00
Research area × Degree	4	.09	.23	1.08	.37	.00
Perceived integrity (RQ2)						
Research area	2	.14	.07	.13	.88	.00
Degree of benevolence	**2**	**10.33**	**5.17**	**9.65**	**.00**	**.05**
Research area × Degree	4	.88	.22	.41	.80	.00

Note: Research area refers to three possible scientist biographies under each condition (crop genetics, corn and soy genetics, or biotechnology use) survey respondents were presented with before responding to questions about perception. Degree of benevolence could be high, with no specific information, or low with conditions based on pretests of information that was likely to communicate high or low degrees of benevolence. Bold font indicates significance ($p < .05$).
SS: sum of squares; *MS*: mean squares.

Perceived benevolence

Consistent with H1, the main effect of degree of benevolence was statistically significant (F (2, 365) = 15.62, $p < .01$) for profiles with a high, unspecified, or low degree of benevolence, with the eta of .08 suggesting a medium effect. As expected, the main effect of research area on perceived benevolence was not significant, F (2, 364) = 1.21, $p = .30$. The interaction effect was also not significant, F (2, 365) = .62, $p = .65$, which suggests that degree of benevolence information influenced participants' reported benevolence perceptions, regardless of condition.

Willingness to trust

Consistent with H2, the main effect of degree of benevolence information was a significant predictor of willingness to trust (F (2, 364) = 5.51, $p < .01$) for profiles with a high, unspecified, or low degree of benevolence. The eta of .03 suggests a small effect. The main effect of research area on willingness to trust was not significant, F (2, 364) = 1.05, $p = .35$, and the interaction effect was not significant, F (2, 364) = .78, $p = .54$.

Perceived competence

Partially answering RQ1, the main effect of degree of benevolence on perceived competence was not significant at F (2, 365) = .47, $p = .62$, for profiles with a high, unspecified, or low degree of benevolence. The interaction effect was not significant at F (2, 365) = 1.080, $p > .05$. The main effect, F (2, 365) = .67, $p = .51$, of research area on perceived competence was not significant.

Perceived integrity

Partially answering RQ2, the main effect of degree of benevolence was significant at F (2, 362) = 9.65, $p < .01$, for profiles with a high, unspecified, or low degree of benevolence. The eta of .05 suggests a medium effect on participants' reported integrity perceptions. The interaction effect was not significant at F (2, 362) = .41, $p = .80$. At F (2, 362) = .13, $p = .88$, the main effect of research area on perceived integrity was not significant.

Overall, our findings suggest that the degree of benevolence in a biography can have a positive impact on perceived benevolence (H1), willingness to trust (H2), and perceived integrity (RQ2) regardless of research area; however, it had no statistically significant effect on perceived competence (RQ1). As the research area had no significant effect on the outcome of any of our variables of interest, we opted to combine participants assigned to the same condition (high, unspecified, low) for our analysis of the main effects and analysis of interaction effects with degree of benevolence.

Direct and indirect effects of degree of benevolence on willingness to trust (H3)

Using Hayes' (2018) PROCESS macro, we tested the direct and indirect effects of degree of benevolence information on trust with perceived benevolence, perceived integrity, and perceived competence as mediators. Our analysis used model 4, which allowed us to input multiple parallel mediators and examine how they mediate the relationship between benevolence information and trust (see Figure 1 and Table 3).

Prior to mediation, our analysis indicated that degree of benevolence information had a significant positive effect on perceived benevolence, willingness to trust, and perceived integrity. The mediation model (Figure 1) suggests that the direct effect of degree of benevolence information on willingness to trust disappears when accounting for trustworthiness mediators, consistent with past theory. More specifically, it appears benevolence perceptions are important for understanding the difference between both high and unspecified degrees of benevolence information, as well as the difference between high and low degrees of benevolence information. High and low degrees of benevolence information also mediated integrity perceptions.

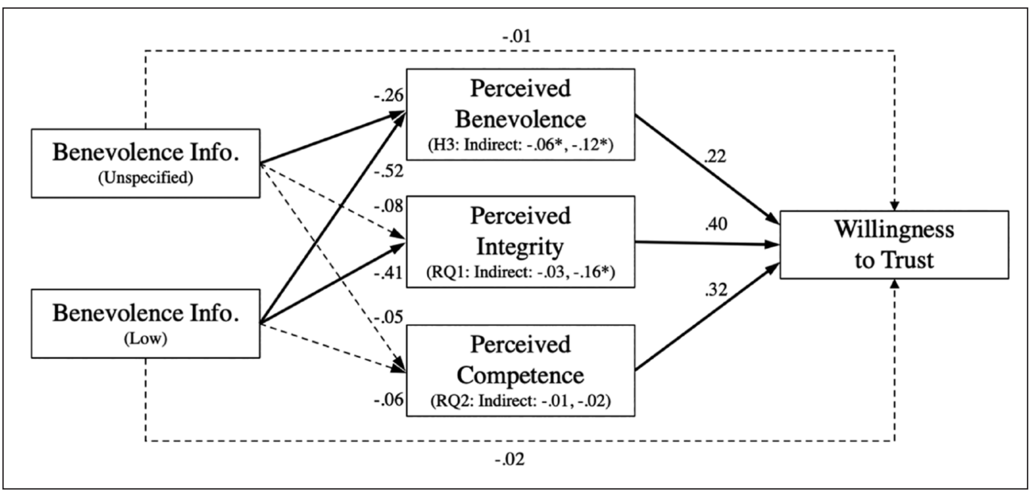

Figure 1. Direct and indirect effects of degree of benevolence on willingness to trust.

Table 3. Standardized direct and indirect effects of degree of benevolence information in scientist profiles.

	β	*SE*	95% CI
Direct effects			
Constant	.11	.35	[−.60, .77]
DoBen (high) → W2Trust (reference)			
DoBen (unspecified) → W2Trust	−.01	.07	[−.15, .13]
DoBen (low) → W2Trust	−.03	.09	[−.19, .15]
DoBen (high) → perceived benevolence (reference)			
DoBen (unspecified) → Perceived benevolence	**−.37**	**.09**	**[−.44, −.09]**
DoBen (low) → Perceived benevolence	**−.71**	**.09**	**[−.68, −.33]**
DoBen (high) → perceived integrity (reference)			
DoBen (unspecified) → Perceived integrity	−.14	.09	[−.28, .08]
DoBen (low) → Perceived integrity	**−.52**	**.10**	**[−.57, −.20]**
DoBen (high) → perceived competence (reference)			
DoBen (unspecified) → Perceived competence	−.10	.06	[−.16, .07]
DoBen (low) → Perceived competence	−.08	.06	[−.15, .08]
Perceived benevolence → W2Trust	**.22**	**.06**	**[.12, .35]**
Perceived integrity → W2Trust	**.38**	**.06**	**[.28, .50]**
Perceived competence → W2Trust	**.19**	**.07**	**[.17, .46]**
Indirect effects			
DoBen (high) → perceived benevolence → W2 T (reference)			
DoBen (unspecified)→ **Perceived benevolence → W2Trust**	**−.08**	**.04**	**[−.16, −.02]**
DoBen (low) → **Perceived benevolence → W2Trust**	**−.15**	**.05**	**[−.27, −.06]**
DoBen (high) → perceived competence → W2 T (reference)			
DoBen (unspecified) → Perceived Comp. → W2Trust	−.02	.02	[−.07, .03]
DoBen (low) → Perceived Comp. → W2Trust	−.02	.02	[−.07, .03]
DoBen (high) → perceived integrity → W2Trust (reference)			
DoBen (unspecified) → Perceived Integ. → W2Trust	−.05	.05	[−.15, .04]
DoBen (low) → Perceived Integ. → W2Trust	**−.20**	**.06**	**[−.32, −.09]**
	R^2	*MSE*	$F_{(5, 367)}$
Model statistics	.44	.33	59.64

Note. Bold values indicate significance ($p < .05$).
SE: standard error; *CI*: confidence interval; *MSE*: mean squared error; DoBen: degree of benevolence; W2Trust: willingness to trust; Perceived Comp.: perceived competence; Perceived Integ.: perceived integrity.

3. Discussion

Our findings suggest that simple strategic communication choices could, at least, yield short-term effects for scientists seeking to improve their trustworthiness in the eyes of others. Biographies can serve as more than just rote pieces of text: They are also a way for scientists to convey their trustworthiness. Conversely, failing to include information on prosocial motivations could invite people to see scientists as less benevolent. This is consistent with existing literature, such as communicating benevolence-type cues including prosocial quotes and non-verbal cues can foster perceptions of “warmth” in scientists (Barger and Grandey, 2006; Zahry Nagwan and Besley, 2021a). In the clinical context, "caring" behaviors and "commitment" messages can enhance perceived likability and the positive view of healthcare providers (Sharp et al., 1992; Turner and Choung, 2023).

Consistent with the integrative model of organizational trust, including information about prosocial motivations in scientist biographies appears to, at least temporarily, increase willingness to trust as a function of its effect on benevolence perceptions, as well as integrity perceptions. Benevolence information affecting integrity perceptions is not especially surprising, as benevolence and integrity perceptions are often highly correlated (Besley et al., 2021) or even considered facets of a singular construct (Choung et al., 2023).

While we observed no differences in perceived competence across conditions, we did note differences in perceived integrity between the high benevolence and non-specified conditions. This variation highlights the possibility that scientific communicators can convey information that fosters benevolence perceptions (i.e. prosocial motivations) without concurrently presenting information explicitly aimed at enhancing perceptions of integrity. In addition, our "low benevolence" biographies, which highlighted scientists' industry connections, necessitate further reflection. Prior research, including studies by Besley et al. (2019), indicates that any connection to industry can significantly invoke conflict of interest perceptions. Future studies could explore whether such portrayals of scientists' work can intensify sensitivity to potential conflicts and financial motivations, thereby influencing integrity perception.

In interpreting the results of our study, we are cognizant of its many limitations. The most evident limitation is the generalizability of our findings. Our study was limited to a specific means of communication (i.e. biographies) and used an online sample that cannot be generalized to the larger population. We also recognize the complexities inherent in translating observed short-term to real-world, long-term attitudinal changes, particularly within the complex landscape of scientific communication teeming with diverse, often competing narratives. While our findings, highlighting the efficacy of benevolence communication in enhancing scientist trustworthiness, are promising, they serve primarily as an initial step. The practical implementation of emphasizing benevolence in scientific discourse will require strategic, context-specific approaches aimed at mitigating the influence of conflicting narratives and at reinforcing the impact and endurance of benevolent scientific messages in the dynamic informational environment. Moreover, long-term trust-building through strategic communication necessitates sustained and consistent efforts to manifest trustworthiness across diverse platforms, extending beyond isolated messages. To form perceptions of scientists as trustworthy, people must be regularly exposed to examples of trustworthy scientists and have opportunities to attend to scientists' trustworthy characteristics, echoing the findings of other communication effects research (Slater and Rasinski, 2005).

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Supplemental material

Supplemental material for this article is available online.

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